

halox: Dark matter halo properties and large-scale structure calculations using JAX

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Software

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Summary

Dark matter halos are fundamental structures in cosmology, forming the gravitational potential wells hosting galaxies and clusters of galaxies. Their properties and statistical distribution (including the halo mass function) are invaluable tools to infer the fundamental properties of the Universe. The `halox` package is a JAX-powered Python library enabling differentiable and accelerated computations of key properties of dark matter halos, and of the halo mass function. The automatic differentiation capabilities of `halox` enable its usage in gradient-based workflows, e.g. in efficient Hamiltonian Monte Carlo sampling or machine learning applications. The acceleration capabilities of `halox` enable significant speedups over existing packages such as `colossus` on GPU architectures, while offering comparable performance on CPUs.

Statement of need

In cosmology and astrophysics, modeling dark matter halos is central to understanding the large-scale structure of the Universe and its formation. Recently, the AI-driven advent of novel computational frameworks such as JAX ([Bradbury et al., 2018](#)) has led to the development of differentiable and hardware-accelerated software to simulate and model physical processes, with e.g. `Brax` ([Freeman et al., 2021](#)) and `JAX, MD` ([Schoenholz & Cubuk, 2020](#)). The increasing complexity of cosmological data and astrophysical models has motivated the wide adoption of this framework in cosmology, where JAX-powered software has been published to address a wide variety of scientific goals, including modeling fundamental cosmological quantities, with, e.g., `JAX-cosmo` ([Campagne et al., 2023](#)) and `LINX` ([Giovanetti et al., 2024](#)); simulating density fields and observables, with, e.g., `SHAMNet` ([Hearin et al., 2022](#)), `DISCO-DJ` ([Hahn et al., 2024](#)), `JAXpm` ([Differentiable Universe Initiative, 2025](#)), and `JAX-GalSim` ([JAX-GalSim developers, 2025](#); [Mendoza et al., 2025](#)); emulating likelihoods for accelerated inference, with, e.g., `CosmoPower-JAX` ([Piras & Spurio Mancini, 2023](#)) and `candl` ([Balkenhol et al., 2024](#)); or modeling various physical properties of dark matter halos, such as mass accretion history (`Diffmah`, [Hearin et al., 2021](#)), galaxy star formation history (`Diffstar`, [Alarcon et al., 2023](#)), halo concentration (`Diffprof`, [Stevanovich et al., 2023](#)), gas-halo connection (`picasso`, [Kéruzoré et al., 2024](#)), and halo mass function ([Buisman et al., 2025](#))¹.

The `halox` library offers a JAX implementation of some widely used properties of dark matter halos. The use of JAX as a backend allows these functions to be compiled and GPU-accelerated, enabling high-performance computations; and automatically differentiable, enabling their efficient use in gradient-based workflows, such as sensitivity analyses, Hamiltonian Monte-Carlo sampling for Bayesian inference, or machine learning-based methods. In addition,

¹Note that `halox` also provides an implementation of the halo mass function, but chooses a lighter, halo model-based approach; see [Software design](#).

expensive computations of large-scale structure properties are further accelerated using neural network emulators, preserving hardware acceleration and differentiability while enabling faster calculations thanks to approximate calculations (see the **Software design** section).

State of the field

Many toolkits focused on halo modeling have been developed, such as, e.g., halofit (Smith et al., 2003), halotools (Hearin et al., 2017), colossus (Diemer, 2018), and pyCCL (Chisari et al., 2019). While JAX-powered software exists for a wide variety of cosmological applications (see **Statement of need**), the properties implemented in halox (listed in **Software design**), although available in other libraries focused on halo modeling, do not currently have publicly available, differentiable and hardware-accelerated implementations.

Software design

Available physical quantities

The halox library seeks to provide JAX-based implementations of common models of dark matter halo properties and of large-scale structure. At the time of writing (software version 2.1.1), this includes the following properties:

- Cosmological quantities: halox relies on JAX-cosmo (Campagne et al., 2023) for cosmology-dependent calculations, and includes wrapper functions to compute some additional properties, such as critical density ρ_c and differential comoving volume element $dV_c/d\Omega dz$.
- Radially-dependent physical properties of NFW and Einasto dark matter halos. Our NFW and Einasto implementations are based on the analytical derivations of Łokas & Mamon (2001) and Retana-Montenegro et al. (2012) respectively, and include the following quantities:
 - Matter density $\rho(r)$;
 - Enclosed mass $M(\leq r)$;
 - Gravitational potential $\phi(r)$;
 - Circular velocity $v_{\text{circ}}(r)$;
 - Velocity dispersion $\sigma_v(r)$ (NFW only);
 - Projected surface density $\Sigma(r)$ (NFW only).
- Concentration-mass relations: There are implementations of several relations including:
 - Duffy et al. (2008)
 - Klypin et al. (2011)
 - Prada et al. (2012)
 - Child et al. (2018) (for all halo and relaxed halo populations)
- Large-scale structure: Building upon the power spectra computations implemented in JAX-cosmo, halox provides implementations of the RMS variance of the matter distribution in spheres of radius R , $\sigma(R)$. It also includes a wrapper function to perform the computation within the Lagrangian radius of a halo of mass M , $\sigma(M)$.
- Halo mass function (HMF): The HMF model of Tinker et al. (2008), predicting $dN/d\ln M$ as a function of halo mass M , redshift z , and cosmology.
- Halo bias: The linear bias model of Tinker et al. (2010) as a function of halo mass M , redshift z , and cosmology.
- Overdensities: All properties in halox can be computed for spherical overdensity (SO) halo masses defined for any critical overdensity value. Convenience functions are provided to convert halo properties from one critical overdensity to another, or to convert critical overdensities to and from mean matter overdensities.

Automatic differentiation and hardware acceleration

All calculations available in `halox` are written using JAX and JAX-cosmo. As a result, all functions can be compiled just-in-time using `jax.jit`, hardware-accelerated, and are automatically differentiable with respect to their input parameters, including halo mass, redshift, and cosmological parameters. In addition, all JAX transformations can be used on `halox` functions, including native vectorization and parallelization using e.g. `jax.vmap`.

Emulation

$\sigma^2(R)$ is the variance of the fluctuations of the matter density field in a sphere of radius R . At redshift z and for cosmological parameters Ω , it is given by:

$$\sigma^2(R, z, \Omega) = \frac{1}{2\pi^2} \int_0^\infty k^2 W^2(k, R) P(k, z, \Omega) dk,$$

where k denotes spatial frequency, $P(k, z, \Omega)$ is the power spectrum, and W is the Fourier transform of the spherical top-hat window function. σ is an essential ingredient in computing both halo mass function and halo bias in most standard parameterizations (e.g., Tinker et al., 2010), and the numerical integration is computationally expensive, and often the primary bottleneck in such calculations.

To tackle this issue, `halox` also includes an emulated calculation of σ , as a function of mass (the Lagrangian mass contained in a radius R), redshift, and cosmological parameters. Our emulator consists of a multi-layer perceptron with three hidden layers, each of width 64. The emulator was trained on the `halox` $\sigma(M)$ implementation. The training set is taken from a Sobol sample over $\log(M)$, $\log(1+z)$, and the cosmological parameters Ω_b , Ω_c , h , n_s , and σ_8 .

The emulator is accurate to within a percent for both $\sigma(M)$ and the halo bias, and within a few percent for the HMF across the tested parameter space. Figure 1 and Figure 2 show the accuracy of emulator-based predictions of $\sigma(M)$ and of the halo mass function for different cosmologies up to $z = 1$, demonstrating remarkable accuracy, with an increase of error in the regime of extremely massive halos ($M_{200c} > 10^{15} h^{-1} M_\odot$), which are very rare occurrences in both simulations and observations.

To compute $\sigma(M)$, the HMF, or the halo bias using the emulator, users may simply instantiate the emulator, then pass it in as an optional argument to the original $\sigma(M)$ function in `halox`:

```
emu = emus.sigmaM.SigmaMEmulation()           # instantiate emulator
sigma_a = halox.lss.sigmaM(M, z, cosmo)        # analytic sigma(M)
sigma_e = halox.lss.sigmaM(M, z, cosmo, emu=emu) # emulated sigma(M)
```

The same calling sequence can be used to compute halo mass function or halo bias using the $\sigma(M)$ emulator.

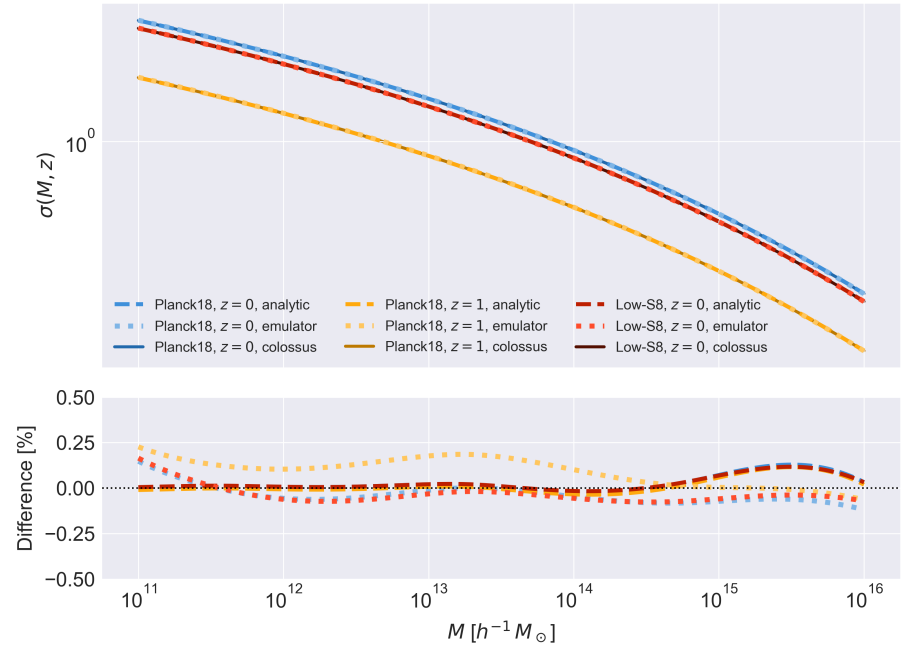


Figure 1: *Top:* $\sigma(M)$ predicted using colossus, halox, and the halox emulator, varying redshift and cosmology. *Bottom:* Fractional difference between halo predictions and colossus.

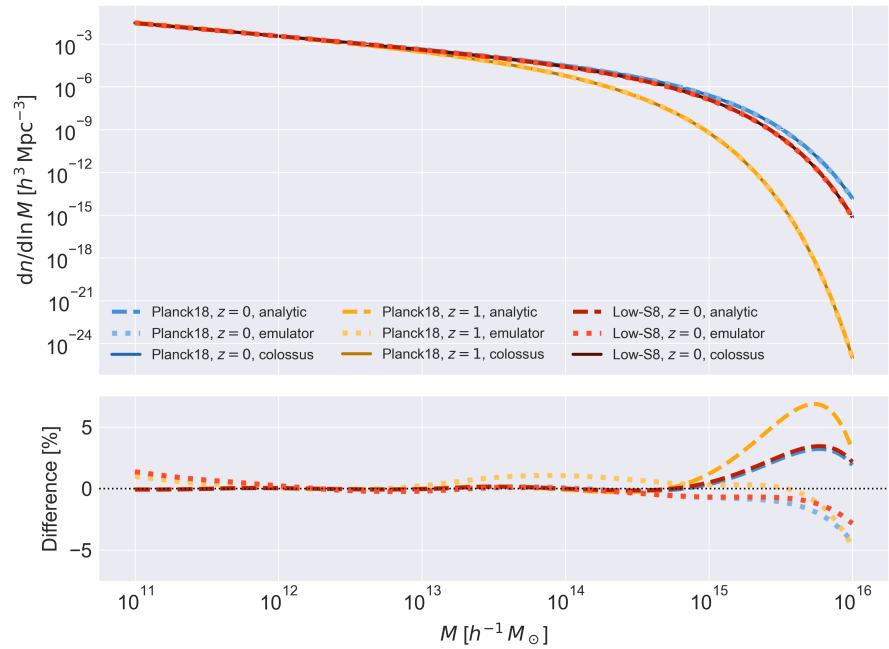


Figure 2: Same as Figure 1 for the halo mass function.

Research impact statement

At the time of writing, `halox` is being used in several upcoming analyses, including the development of a differentiable generator of halo catalogs (Zacharegkas et al., 2026) and the creation of a suite of realistic simulations of the extragalactic millimeter-wave sky (Kéruzoré et al., 2026). In addition, it has already been referenced in cosmology publications (e.g., LSST Dark Energy Science Collaboration et al., 2026).

Performance benchmarks

Figure 3 shows the performance of halo mass function computations in `halox` across hardware configurations, benchmarked against our slowest evaluation (analytical computation on CPU). For comparison, we also compare to the performance of `colossus` on the same computation². Three results stand out. First, we see that `colossus` outperforms `halox` on CPU, by about a factor of three for the analytic computation, owing to calls to highly efficient libraries such as CAMB (Lewis & Challinor, 2011) and well-optimized interpolation and integration schemes. Second, using a GPU significantly accelerates `halox` predictions, by a factor of over 20 for the analytic computation, and of about 65 for the emulated version. Third, using the neural network emulator as a backend for the $\sigma(M)$ computation enables a substantial speedup, up to 95 $\times$ compared to baseline, and 34 $\times$ compared to `colossus`. These results demonstrate the strong potential of `halox` in GPU-based cosmological analyses, delivering considerable speedup in addition to automatic differentiation. We also note that all computations were made at double (FP64) precision; `halox` can be further accelerated, in particular on GPUs, by dropping to single (FP32) precision.

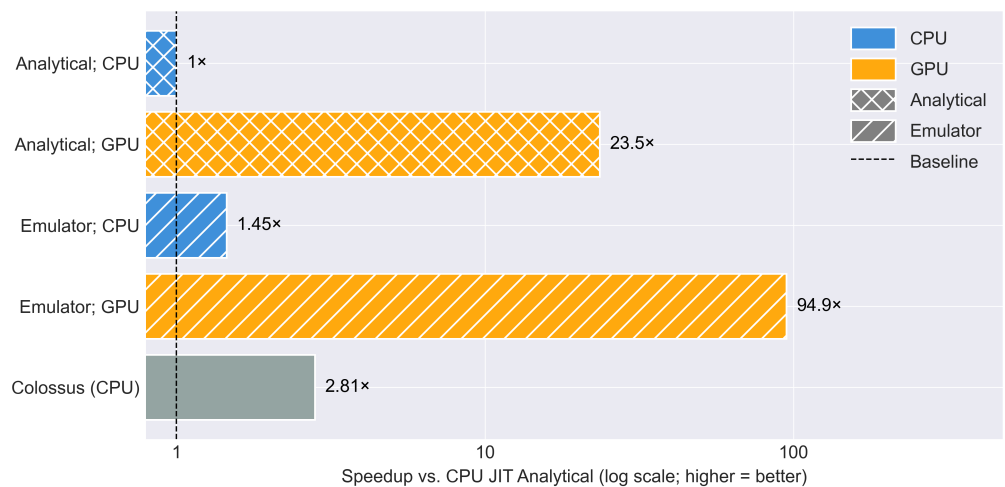


Figure 3: The performance of HMF computation for the `halox` package on different architectures and against `colossus`. All CPU executions are still slower than `colossus` irrespective of emulation. GPU architecture enables further speedup, allowing for faster computations than `colossus` both with and without emulation, with significant speedup when using the emulated function over the standard calculation.

Validation

All functions available in `halox` are validated against existing, non-JAX-based software. Cosmology calculations are validated against `Astropy` (Astropy Collaboration et al., 2022) for

²Our benchmarks were run on an AMD EPYC 7742 CPU (2.25GHz) and an NVIDIA A100-SXM4-40GB GPU, evaluating the HMF on a grid of 256 halo masses $\times$ 256 redshifts, for a fixed cosmology. All `halox` computations were run after just-in-time compilation.

139 varying cosmological parameters and redshifts. Other quantities are validated against either
140 colossus (Diemer, 2018) or Gala (Price-Whelan, 2017) for varying halo masses, redshifts,
141 critical overdensities, and cosmological parameters. These tests are included in an automatic
142 CI/CD pipeline on the GitHub repository, and presented graphically in the online documentation.

143 AI usage disclosure

144 We acknowledge the use of Anthropic's Claude (Sonnet 4.5, Opus 4.5, Opus 4.6) in the
145 development of halox (code and documentation). We note that no AI was used in writing
146 the unit test suite enforcing accuracy of the software, which allowed us to ensure that the
147 generated code produced correct results.

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